

The Analytic Structure of Quadratic Normal Volatility Models: From the Black–Scholes PDE to the Confluent Heun Equation

Abstract

We derive a canonical mapping from the Quadratic Normal Volatility (QNV) local-volatility specification to the confluent Heun equation, starting from the Black–Scholes partial differential equation and proceeding through time reversal, Lamperti reparametrization, and a gauge scaling that yields a one-dimensional Schrödinger form with computable potential. The resulting second-order ordinary differential equation admits solutions within the Heun family, enabling structural analysis, asymptotics, and polynomial-termination regimes under explicit algebraic constraints. This analytic route clarifies why QNV achieves tractability while preserving a purely local (one-factor) market geometry, and it exposes precise limitations for forward-smile dynamics. The outline declares the symbolic pipeline, parameter extraction, and rigorous conditions for Heun confluent structure, with a focus on exact operator identities and lawful transformations (no numerics). Emphasis is placed on uniform equation referencing, invariance under numeraire/gauge changes, and conservative use of proof sketches anchored by well-established literature.

1 Motivation, Model Landscape, and QNV Specification

1.1 Empirical Motivations and the Volatility Smile

The foundational work of [2] and [17] established option pricing theory on the assumption of geometric Brownian motion with constant volatility. This framework yields the celebrated Black–Scholes formula and the concept of implied volatility as a market language. However, post-1987 crash markets systematically exhibit *volatility smiles* and *skews*—patterns in implied volatility across strikes that contradict the constant-volatility assumption [7]. These patterns indicate risk-neutral distributions with heavier tails and skewness than the log-normal distribution.

This empirical reality motivated three primary modeling approaches:

1. **Local Volatility (LV) Models:** Specify volatility as a deterministic function $\sigma(S, t)$, perfect for calibrating to current option prices [5, 4].
2. **Stochastic Volatility (SV) Models:** Model volatility as a separate stochastic process, capturing realistic dynamics [9].
3. **Jump-Diffusion Models:** Add discontinuous jumps to returns, capturing tail events [10].

The Quadratic Normal Volatility (QNV) model is a parametric LV model notable for its blend of flexibility and analytical tractability. Its volatility function is:

$$\sigma(S) = aS^2 + bS + c \tag{1}$$

where a controls smile convexity, b controls skew, and c sets the base volatility level. The discriminant $\Delta = b^2 - 4ac$ determines the root structure and smile shape.

1.2 Financial PDE and Canonical Transformations

The Black-Scholes PDE for a derivative price $C(S, t)$ is:

$$\frac{\partial C}{\partial t} + \frac{1}{2}\sigma^2(S)S^2\frac{\partial^2 C}{\partial S^2} + rS\frac{\partial C}{\partial S} - rC = 0 \quad (2)$$

Substituting the QNV form (1) yields a PDE with a quartic diffusion coefficient. We simplify through financial and mathematical transformations.

First, apply time reversal and discounting:

$$\tau = T - t \quad (3)$$

$$v(S, \tau) = e^{r\tau}C(S, t) \quad (4)$$

Transforming (2) gives the forward equation:

$$\frac{\partial v}{\partial \tau} = \frac{1}{2}\sigma^2(S)\frac{\partial^2 v}{\partial S^2} + rS\frac{\partial v}{\partial S} \quad (5)$$

Second, apply the Lamperti transform to remove state-dependent diffusion [14, 16]:

$$x(S) = \int \frac{dS}{\sigma(S)} = \int \frac{dS}{aS^2 + bS + c} \quad (6)$$

This integral has closed forms depending on Δ :

$$x(S) = \begin{cases} \frac{2}{\sqrt{\Delta}} \operatorname{arctanh}\left(\frac{2aS+b}{\sqrt{\Delta}}\right) & \Delta > 0 \\ -\frac{2}{2aS+b} & \Delta = 0 \\ \frac{2}{\sqrt{-\Delta}} \operatorname{arctan}\left(\frac{2aS+b}{\sqrt{-\Delta}}\right) & \Delta < 0 \end{cases} \quad (7)$$

For $\Delta > 0$ (hyperbolic case), define:

$$t = \tanh\left(\frac{\sqrt{\Delta}}{2}x\right) = \frac{2aS+b}{\sqrt{\Delta}} \quad (8)$$

inverting to:

$$S(t) = \frac{-b + \sqrt{\Delta}t}{2a} \quad (9)$$

Using the chain rule, the spatial derivatives become:

$$\frac{\partial}{\partial S} = \frac{1}{\sigma(S)} \frac{\partial}{\partial x} \quad (10)$$

$$\frac{\partial^2}{\partial S^2} = -\frac{\sigma'(S)}{\sigma^2(S)} \frac{\partial}{\partial x} + \frac{1}{\sigma^2(S)} \frac{\partial^2}{\partial x^2} \quad (11)$$

Substituting into (5) yields:

$$\frac{\partial v}{\partial \tau} = \frac{1}{2} \frac{\partial^2 v}{\partial x^2} + \mu(x) \frac{\partial v}{\partial x} \quad (12)$$

where the drift function is:

$$\mu(x) = \frac{rS(x)}{\sigma(S(x))} - \frac{1}{2}\sigma'(S(x)) \quad (13)$$

Equation (12) has constant diffusion but state-dependent drift. This sets the stage for the gauge transformation that will complete the reduction to Schrödinger form, revealing the connection to the confluent Heun equation [3, 21]. The analytical tractability of QNV stems from this exact reducibility, contrasting with more complex models requiring numerical methods.

2 Lamperti Transformation and Drift Neutralization

2.1 Lamperti Transformation and Coordinate Change

The forward PDE derived in equation (12) features constant diffusion but retains a state-dependent drift term $\mu(x)$. To eliminate this drift and reveal the underlying Schrödinger structure, we employ a gauge transformation following the approach of [14, 16].

We begin with the transformed PDE:

$$\frac{\partial v}{\partial \tau} = \frac{1}{2} \frac{\partial^2 v}{\partial x^2} + \mu(x) \frac{\partial v}{\partial x} \quad (14)$$

Define a gauge transformation:

$$v(x, \tau) = g(x) \psi(x, \tau) \quad (15)$$

Substitute into (14):

$$\frac{\partial v}{\partial \tau} = g(x) \frac{\partial \psi}{\partial \tau} \quad (16)$$

$$\frac{\partial v}{\partial x} = g'(x) \psi + g(x) \frac{\partial \psi}{\partial x} \quad (17)$$

$$\frac{\partial^2 v}{\partial x^2} = g''(x) \psi + 2g'(x) \frac{\partial \psi}{\partial x} + g(x) \frac{\partial^2 \psi}{\partial x^2} \quad (18)$$

Substituting into (14):

$$g \frac{\partial \psi}{\partial \tau} = \frac{1}{2} \left(g'' \psi + 2g' \frac{\partial \psi}{\partial x} + g \frac{\partial^2 \psi}{\partial x^2} \right) + \mu \left(g' \psi + g \frac{\partial \psi}{\partial x} \right) \quad (19)$$

Divide by g and rearrange:

$$\frac{\partial \psi}{\partial \tau} = \frac{1}{2} \frac{\partial^2 \psi}{\partial x^2} + \left(\frac{g'}{g} + \mu \right) \frac{\partial \psi}{\partial x} + \left(\frac{g''}{2g} + \mu \frac{g'}{g} \right) \psi \quad (20)$$

To eliminate the first-order term, we choose g such that:

$$\frac{g'}{g} + \mu = 0 \quad \Rightarrow \quad g'(x) = -\mu(x)g(x) \quad (21)$$

This yields the solution:

$$g(x) = \exp \left(- \int \mu(x) dx \right) \quad (22)$$

With this choice, the PDE simplifies to:

$$\frac{\partial \psi}{\partial \tau} = \frac{1}{2} \frac{\partial^2 \psi}{\partial x^2} + V(x) \psi \quad (23)$$

where the potential function is:

$$V(x) = \frac{1}{2} \left(\mu^2(x) + \mu'(x) \right) \quad (24)$$

2.2 Separation of Variables and Stationary Equation

Apply separation of variables:

$$\psi(x, \tau) = e^{-E\tau} \phi(x) \quad (25)$$

Substituting into (23) yields the time-independent Schrödinger equation:

$$-\frac{1}{2} \frac{d^2 \phi}{dx^2} + V(x) \phi = E \phi \quad (26)$$

or equivalently:

$$\left[-\frac{1}{2} \frac{d^2}{dx^2} + V(x) \right] \phi(x) = E \phi(x) \quad (27)$$

3 Schrödinger Reduction and Potential Extraction

3.1 Explicit Form of the Potential

To proceed further, we need the explicit form of the potential $V(x)$ in terms of the original parameters. From (24), we have:

$$V(x) = \frac{1}{2} \left(\mu^2(x) + \mu'(x) \right) \quad (28)$$

Recall the drift function:

$$\mu(x) = \frac{rS(x)}{\sigma(S(x))} - \frac{1}{2}\sigma'(S(x)) \quad (29)$$

For the QNV specification $\sigma(S) = aS^2 + bS + c$, we compute the derivative:

$$\sigma'(S) = 2aS + b \quad (30)$$

Using the hyperbolic coordinate transformation (8) for the case $\Delta > 0$:

$$t = \tanh\left(\frac{\sqrt{\Delta}}{2}x\right) = \frac{2aS + b}{\sqrt{\Delta}} \quad (31)$$

We can express all quantities in terms of t :

$$S(t) = \frac{-b + \sqrt{\Delta}t}{2a} \quad (32)$$

$$\sigma(S(t)) = \frac{\Delta}{4a}(t^2 - 1) \quad (33)$$

$$\sigma'(S(t)) = \sqrt{\Delta}t \quad (34)$$

$$\frac{dt}{dx} = \frac{\sqrt{\Delta}}{2}(1 - t^2) \quad (35)$$

Substituting these into the drift function:

$$\mu(x) = \frac{rS}{\sigma(S)} - \frac{1}{2}\sigma'(S) \quad (36)$$

$$= \frac{r \left(\frac{-b + \sqrt{\Delta}t}{2a} \right)}{\frac{\Delta}{4a}(t^2 - 1)} - \frac{1}{2}(\sqrt{\Delta}t) \quad (37)$$

$$= \frac{2r(-b + \sqrt{\Delta}t)}{\Delta(t^2 - 1)} - \frac{\sqrt{\Delta}t}{2} \quad (38)$$

Now compute $\mu'(x) = \frac{d\mu}{dx} = \frac{d\mu}{dt} \frac{dt}{dx}$:

$$\frac{d\mu}{dt} = \frac{d}{dt} \left[\frac{2r(-b + \sqrt{\Delta}t)}{\Delta(t^2 - 1)} \right] - \frac{\sqrt{\Delta}}{2} \quad (39)$$

$$= \frac{2r}{\Delta} \cdot \frac{\sqrt{\Delta}(t^2 - 1) - (-b + \sqrt{\Delta}t) \cdot 2t}{(t^2 - 1)^2} - \frac{\sqrt{\Delta}}{2} \quad (40)$$

$$= \frac{2r}{\Delta} \cdot \frac{\sqrt{\Delta}t^2 - \sqrt{\Delta} + 2bt - 2\sqrt{\Delta}t^2}{(t^2 - 1)^2} - \frac{\sqrt{\Delta}}{2} \quad (41)$$

$$= \frac{2r}{\Delta} \cdot \frac{-\sqrt{\Delta}t^2 + 2bt - \sqrt{\Delta}}{(t^2 - 1)^2} - \frac{\sqrt{\Delta}}{2} \quad (42)$$

Then:

$$\mu'(x) = \left[\frac{2r}{\Delta} \cdot \frac{-\sqrt{\Delta}(t^2 + 1) + 2bt}{(t^2 - 1)^2} - \frac{\sqrt{\Delta}}{2} \right] \cdot \frac{\sqrt{\Delta}}{2}(1 - t^2) \quad (43)$$

$$= \frac{r}{\sqrt{\Delta}} \cdot \frac{-\sqrt{\Delta}(t^2 + 1) + 2bt}{(t^2 - 1)^2} \cdot (1 - t^2) - \frac{\Delta}{4}(1 - t^2) \quad (44)$$

After extensive algebra, we obtain the exact potential:

$$V(t) = \frac{N(t)}{D(t)} \quad (45)$$

where:

$$N(t) = 2\Delta^3 t^6 + (-4r\Delta^2 - 5\Delta^3)t^4 + (16r^2\Delta + 8r\Delta^2 + 4\Delta^3)t^2 \quad (46)$$

$$- 32br^2\sqrt{\Delta}t + (16b^2r^2 - 4r\Delta^2 - \Delta^3) \quad (47)$$

$$D(t) = 8\Delta^2(1 - t^2)^2 \quad (48)$$

3.2 Schrödinger Equation in Hyperbolic Coordinates

The Schrödinger equation (26) becomes:

$$-\frac{1}{2} \frac{d^2\phi}{dx^2} + V(t)\phi = E\phi \quad (49)$$

Using the chain rule:

$$\frac{d\phi}{dx} = \frac{dt}{dx} \frac{d\phi}{dt} = \frac{\sqrt{\Delta}}{2}(1 - t^2) \frac{d\phi}{dt} \quad (50)$$

$$\frac{d^2\phi}{dx^2} = \frac{d}{dx} \left(\frac{\sqrt{\Delta}}{2}(1 - t^2) \frac{d\phi}{dt} \right) \quad (51)$$

$$= \frac{\sqrt{\Delta}}{2} \left(-2t \frac{dt}{dx} \frac{d\phi}{dt} + (1 - t^2) \frac{d}{dx} \frac{d\phi}{dt} \right) \quad (52)$$

$$= \frac{\sqrt{\Delta}}{2} \left(-2t \cdot \frac{\sqrt{\Delta}}{2}(1 - t^2) \frac{d\phi}{dt} + (1 - t^2) \cdot \frac{\sqrt{\Delta}}{2}(1 - t^2) \frac{d^2\phi}{dt^2} \right) \quad (53)$$

$$= \frac{\Delta}{4} \left(-2t(1 - t^2) \frac{d\phi}{dt} + (1 - t^2)^2 \frac{d^2\phi}{dt^2} \right) \quad (54)$$

Substituting into the Schrödinger equation:

$$-\frac{\Delta}{8} \left[(1 - t^2)^2 \frac{d^2\phi}{dt^2} - 2t(1 - t^2) \frac{d\phi}{dt} \right] + V(t)\phi = E\phi \quad (55)$$

Multiply through by $8/\Delta$:

$$-(1 - t^2)^2 \frac{d^2\phi}{dt^2} + 2t(1 - t^2) \frac{d\phi}{dt} + \frac{8}{\Delta}(V(t) - E)\phi = 0 \quad (56)$$

Rearranging gives the final form:

$$(1 - t^2)^2 \frac{d^2\phi}{dt^2} - 2t(1 - t^2) \frac{d\phi}{dt} + \frac{8}{\Delta}(E - V(t))\phi = 0 \quad (57)$$

This equation, with the rational potential $V(t)$, will be transformed to the confluent Heun equation in the next section, revealing the deep mathematical structure underlying the QNV model [19, 3].

4 Confluent Heun Normal Form and Structural Consequences

4.1 Transformation to Confluent Heun Equation

The derived equation (57) represents a second-order differential equation with regular singular points at $t = \pm 1$. To transform this into the canonical confluent Heun form, we follow the established procedure from [18, 20, 11, 13].

We begin with the equation in hyperbolic coordinates:

$$(1 - t^2)^2 \frac{d^2 \phi}{dt^2} - 2t(1 - t^2) \frac{d\phi}{dt} + \frac{8}{\Delta} (E - V(t)) \phi = 0 \quad (58)$$

First, we factor out the singular behavior at the regular singular points $t = \pm 1$. The indicial exponents at these points are found by substituting $\phi(t) \sim (1 \mp t)^\rho$ and solving for ρ . This yields:

$$\alpha = 1 + \frac{r(b - \sqrt{\Delta})}{a\sqrt{\Delta}} \quad (59)$$

$$\beta = 1 - \frac{r(b + \sqrt{\Delta})}{a\sqrt{\Delta}} \quad (60)$$

We therefore make the ansatz:

$$\phi(t) = (1 - t)^\alpha (1 + t)^\beta f(t) \quad (61)$$

Next, we change variables to map the singular points to $y = 0$ and $y = 1$:

$$y = \frac{1 - t}{2} \quad \Leftrightarrow \quad t = 1 - 2y \quad (62)$$

The derivatives transform as:

$$\frac{d}{dt} = -\frac{1}{2} \frac{d}{dy} \quad (63)$$

$$\frac{d^2}{dt^2} = \frac{1}{4} \frac{d^2}{dy^2} \quad (64)$$

Substituting (61) and the variable change into the differential equation, after extensive algebra we obtain the canonical confluent Heun equation:

$$\frac{d^2 f}{dy^2} + \left(\gamma + \frac{\delta}{y} + \frac{\epsilon}{y - 1} \right) \frac{df}{dy} + \frac{\alpha_H y - q}{y(y - 1)} f = 0 \quad (65)$$

The parameters of the confluent Heun equation are explicitly given by:

$$\gamma = -\frac{4r}{\sqrt{\Delta}} \quad (66)$$

$$\delta = 1 + \frac{2r(b - \sqrt{\Delta})}{a\sqrt{\Delta}} \quad (67)$$

$$\epsilon = 1 + \frac{2r(b + \sqrt{\Delta})}{a\sqrt{\Delta}} \quad (68)$$

$$\alpha_H = \frac{8}{\Delta} \left(E - \frac{r}{2} - \frac{\Delta}{8} \right) \quad (69)$$

$$q = \frac{4}{\Delta} \left[E - \frac{r}{2} - \frac{\Delta}{8} \left(1 - \frac{2r(b - \sqrt{\Delta})}{a\sqrt{\Delta}} \right) - \frac{2r^2}{\Delta} - \frac{2br\sqrt{\Delta}}{4a} \right] \quad (70)$$

A crucial observation is the asymmetry in the parameters:

$$\delta - \epsilon = -\frac{4r}{a} \quad (71)$$

This asymmetry vanishes only when $r = 0$ (no interest rates) or $a \rightarrow \infty$ (degenerate case), confirming it as a fundamental feature of the QNV model with nonzero interest rates.

4.2 Structural Consequences and Exact Regimes

The reduction to the confluent Heun equation (65) reveals deep mathematical structure and enables analytical progress through known properties of Heun functions [6, 18, 20].

4.2.1 Indicial Analysis and Singularity Structure

The confluent Heun equation has regular singular points at $y = 0$, $y = 1$, and an irregular singular point at infinity. The indicial exponents at the regular singular points are:

At $y = 0$:

$$\rho_{1,2} = 0, 1 - \delta \quad (72)$$

At $y = 1$:

$$\rho_{1,2} = 0, 1 - \epsilon \quad (73)$$

These exponents determine the local behavior of solutions near the singular points. The condition for the existence of logarithmic solutions is that the difference between exponents is an integer:

$$\delta \in \mathbb{Z} \quad \text{or} \quad \epsilon \in \mathbb{Z} \quad (74)$$

These conditions translate to specific relationships between the financial parameters (a, b, c, r) .

4.2.2 Polynomial Solutions and Termination Conditions

A particularly important class of solutions are Heun polynomials, which occur when the infinite series solution terminates. According to [6], polynomial solutions exist when the following conditions are satisfied:

1. $\alpha_H = -n(n + 1 + \gamma + \delta + \epsilon)$ for some integer $n \geq 0$
2. A certain determinant condition involving the parameters vanishes

The first condition translates to a quantization of the energy parameter E :

$$E = \frac{r}{2} + \frac{\Delta}{8} - \frac{\Delta}{8}n(n + 1 + \gamma + \delta + \epsilon) \quad (75)$$

The second condition imposes algebraic constraints on the parameters (a, b, c, r) . When both conditions are satisfied, the solution reduces to a polynomial of degree n , providing exact closed-form solutions to the pricing problem.

4.2.3 Three-Term Recurrence and Series Solutions

The confluent Heun equation admits power series solutions around regular points. Substituting $f(y) = \sum_{k=0}^{\infty} c_k y^k$ into (65) yields a three-term recurrence relation:

$$c_{k+1} = A_k c_k + B_k c_{k-1}, \quad k \geq 0 \quad (76)$$

with $c_{-1} = 0$ and where the coefficients A_k, B_k are rational functions of the parameters $(\gamma, \delta, \epsilon, \alpha_H, q)$.

This recurrence relation provides a computationally efficient method for evaluating Heun functions and thus option prices under the QNV model. The convergence properties of the series are determined by the asymptotic behavior of the recurrence coefficients.

4.2.4 Connection Formulas and Asymptotic Behavior

The connection formulas between solutions at different singular points are known for the confluent Heun equation. These allow the determination of the global behavior of solutions from their local expansions.

As $y \rightarrow \infty$ ($t \rightarrow -\infty$), the solutions exhibit asymptotic behavior:

$$f(y) \sim y^{-\alpha_H} e^{-\gamma y} \left[1 + \mathcal{O}\left(\frac{1}{y}\right) \right] \quad (77)$$

This asymptotic behavior is crucial for ensuring that solutions satisfy appropriate boundary conditions in the pricing problem.

The reduction to confluent Heun form thus provides not only a classification of the mathematical structure but also practical computational tools and exact solutions in special cases, explaining the remarkable analytical tractability of the QNV model.

5 Reduction to Pöschl-Teller Potential in the Symmetric Case

5.1 Algebraic Reduction to Exactly Solvable Form

A remarkable property of the Quadratic Normal Volatility (QNV) model emerges in the symmetric case when the risk-free interest rate vanishes ($r = 0$). Under this condition, the general confluent Heun equation reduces to a form described by the Pöschl-Teller potential, a well-known exactly solvable potential in quantum mechanics [3, 21].

The condition for symmetry in the Heun parameters requires:

$$\delta = \epsilon \quad \Rightarrow \quad -\frac{4r}{a} = 0 \quad \Rightarrow \quad r = 0 \quad (78)$$

With $r = 0$, the drift function simplifies substantially:

$$\mu(x) = -\frac{1}{2}\sigma'(S(x)) \quad (79)$$

For the QNV specification $\sigma(S) = aS^2 + bS + c$, we have:

$$\sigma'(S) = 2aS + b \quad (80)$$

Using the hyperbolic coordinate transformation $t = \tanh\left(\frac{\sqrt{\Delta}}{2}x\right) = \frac{2aS+b}{\sqrt{\Delta}}$ with $\Delta = b^2 - 4ac > 0$, we obtain:

$$\mu(x) = -\frac{1}{2}\sqrt{\Delta}t \quad (81)$$

The derivative $\mu'(x)$ is computed as:

$$\mu'(x) = \frac{d\mu}{dx} = \frac{d\mu}{dt} \frac{dt}{dx} = \left(-\frac{1}{2}\sqrt{\Delta}\right) \cdot \left(\frac{\sqrt{\Delta}}{2}(1-t^2)\right) = -\frac{\Delta}{4}(1-t^2) \quad (82)$$

The quantum potential $V(x) = \frac{1}{2}(\mu^2 + \mu')$ then becomes:

$$V(x) = \frac{1}{2} \left(\frac{\Delta}{4}t^2 - \frac{\Delta}{4}(1-t^2) \right) \quad (83)$$

$$= \frac{1}{2} \cdot \frac{\Delta}{4}(2t^2 - 1) = \frac{\Delta}{8}(2t^2 - 1) \quad (84)$$

Substituting $t = \tanh\left(\frac{\sqrt{\Delta}}{2}x\right)$ and using the identity $\tanh^2 u = 1 - \operatorname{sech}^2 u$, we obtain:

$$V(x) = \frac{\Delta}{8} \left(2 \tanh^2 \left(\frac{\sqrt{\Delta}}{2}x \right) - 1 \right) \quad (85)$$

$$= \frac{\Delta}{8} \left(2(1 - \operatorname{sech}^2 \left(\frac{\sqrt{\Delta}}{2}x \right)) - 1 \right) \quad (86)$$

$$= \frac{\Delta}{8} - \frac{\Delta}{4} \operatorname{sech}^2 \left(\frac{\sqrt{\Delta}}{2}x \right) \quad (87)$$

5.2 Connection to Pöschl-Teller Potential

The resulting potential:

$$V(x) = \frac{\Delta}{8} - \frac{\Delta}{4} \operatorname{sech}^2 \left(\frac{\sqrt{\Delta}}{2}x \right) \quad (88)$$

is precisely of the Pöschl-Teller form [21]. The standard Pöschl-Teller potential is given by:

$$V_{\text{PT}}(x) = -\frac{\lambda(\lambda+1)}{2} \operatorname{sech}^2(x) \quad (89)$$

To see the equivalence, let $u = \frac{\sqrt{\Delta}}{2}x$, then:

$$V(u) = \frac{\Delta}{8} - \frac{\Delta}{4} \operatorname{sech}^2(u) \quad (90)$$

The term $-\frac{\Delta}{4} \operatorname{sech}^2(u)$ matches the Pöschl-Teller form when:

$$-\frac{\Delta}{4} = -\frac{\lambda(\lambda+1)}{2} \Rightarrow \lambda(\lambda+1) = \frac{\Delta}{2} \quad (91)$$

The constant term $\frac{\Delta}{8}$ represents an energy shift that does not affect the functional form of the potential or the structure of the solutions.

5.3 Implications for Exact Solvability

This reduction to the Pöschl-Teller potential explains the remarkable analytical tractability of the QNV model in the symmetric case ($r = 0$) [3]. The Pöschl-Teller potential is known to be exactly solvable in quantum mechanics, with solutions expressible in terms of hypergeometric functions.

The eigenvalues (energy levels) for the Pöschl-Teller potential are given by:

$$E_n = -\frac{1}{2}(n - \lambda)^2, \quad n = 0, 1, 2, \dots < \lambda \quad (92)$$

which translates to specific quantization conditions for the QNV model parameters in the symmetric case.

This mathematical connection provides profound insight into why the QNV model, despite its apparent complexity, admits such elegant analytical treatment. It represents a financial realization of a fundamental quantum mechanical system, with the volatility smile corresponding to the quantum potential landscape [21].

The reduction also demonstrates how the symmetry condition $r = 0$ simplifies the model considerably, transforming it from a general confluent Heun equation to a more restricted but exactly solvable form. This illustrates the fundamental role of interest rates in breaking the symmetry of the volatility smile and creating the rich structure captured by the full confluent Heun equation.

6 Risk-Neutral Symmetries, Analytical Implications, and Limitations

6.1 Risk-Neutral Symmetries and Gauge Invariance

The mathematical structure revealed by the confluent Heun mapping exposes deep connections between financial modeling and fundamental physical principles, particularly gauge theory [15, 12]. The sequence of transformations applied to the Black-Scholes PDE—time reversal, discounting, Lamperti transformation, and gauge transformation—can be interpreted financially as changes of numeraire and measure.

The gauge transformation $v(x, \tau) = g(x)\psi(x, \tau)$ with $g'(x)/g(x) = -\mu(x)$ is mathematically equivalent to a change of numeraire in the financial sense. This transformation eliminates the drift term, reflecting the fundamental principle that economic laws should be invariant under changes of the unit of account. The resulting Schrödinger equation exhibits explicit gauge invariance under the transformation:

$$\phi(x) \rightarrow e^{i\Lambda(x)}\phi(x), \quad V(x) \rightarrow V(x) + \frac{1}{2}\Lambda'(x) \quad (93)$$

for any sufficiently smooth function $\Lambda(x)$.

This gauge symmetry corresponds to the financial invariance under numeraire changes. The parameters of the confluent Heun equation $(\gamma, \delta, \epsilon, \alpha_H, q)$ are gauge invariants—they remain unchanged under valid numeraire transformations. This explains the remarkable tractability of the QNV model: it possesses a hidden symmetry that constrains its dynamics and makes it exactly reducible to a known special function.

The asymmetry parameter $\delta - \epsilon = -4r/a$ has a clear financial interpretation: it represents the fundamental tension between the interest rate r (time value of money) and the quadratic volatility parameter a (smile convexity). This invariant vanishes in the risk-neutral world when $r = 0$, recovering the symmetric case.

6.2 Analytical Implications for Calibration

The reduction to confluent Heun form provides powerful analytical tools for model calibration and pricing [1, 7]. The three-term recurrence relation for the series coefficients enables efficient computation of option prices without numerical PDE methods.

For the confluent Heun function $f(y) = \sum_{k=0}^{\infty} c_k y^k$, the coefficients satisfy:

$$c_{k+1} = A_k c_k + B_k c_{k-1}, \quad k \geq 1 \quad (94)$$

with initial conditions:

$$c_1 = A_0 c_0, \quad c_0 \text{ arbitrary} \quad (95)$$

where:

$$A_k = \frac{(k+1)(k+\gamma) + q}{(k+1)(k+\delta)} \quad (96)$$

$$B_k = \frac{\alpha_H - (k-1)(k+\gamma+\delta+\epsilon-1)}{(k+1)(k+\delta)} \quad (97)$$

This recurrence relation provides a computationally stable method for evaluating option prices. The series converges exponentially for $|y| < 1$, and asymptotic expansions are available for large $|y|$.

In the special case of polynomial solutions (when the termination conditions are satisfied), we obtain quasi-exact pricing formulas. These occur when the parameters satisfy the algebraic conditions:

$$\alpha_H = -n(n+1+\gamma+\delta+\epsilon), \quad n \in \mathbb{Z}^+ \quad (98)$$

and an additional determinant condition vanishes. In these regimes, the option price reduces to a finite polynomial:

$$f(y) = \sum_{k=0}^n c_k y^k \quad (99)$$

providing closed-form solutions that are exact up to machine precision.

The calibration problem thus reduces to finding parameters (a, b, c, r) such that the Heun parameters satisfy market constraints. This can be formulated as an optimization problem in the parameter space, leveraging the analytical expressions for the recurrence coefficients.

6.3 Limitations and Conservative Scope

Despite its analytical elegance, the QNV model inherits the fundamental limitations of local volatility models [9, 8]. The most significant constraint is its one-factor structure, which implies that the volatility smile is completely determined by the current asset price. This leads to unrealistic predictions for forward smile dynamics.

By Gyongy’s theorem [8], any local volatility model can replicate the current implied volatility surface exactly. However, the future smile dynamics are constrained by the deterministic function $\sigma(S, t)$. In particular, the QNV model predicts that the smile will flatten as time evolves, contrary to empirical observations that smile structures tend to persist over time.

Mathematically, this limitation is encoded in the confluent Heun structure itself. The equation:

$$\frac{d^2 f}{dy^2} + \left(\gamma + \frac{\delta}{y} + \frac{\epsilon}{y-1} \right) \frac{df}{dy} + \frac{\alpha_H y - q}{y(y-1)} f = 0 \quad (100)$$

has fixed parameters $(\gamma, \delta, \epsilon, \alpha_H, q)$ that do not evolve stochastically. This contrasts with stochastic volatility models like Heston’s [9], where the volatility itself follows a stochastic process, allowing for richer smile dynamics.

The QNV model should therefore be viewed as a powerful but limited tool—excellent for static calibration and understanding structural relationships, but insufficient for capturing the full dynamics of volatility surfaces. Its true value lies in the analytical insights it provides rather than as a comprehensive market model.

The mapping to confluent Heun functions nevertheless represents a significant theoretical advance, demonstrating that certain financial models possess mathematical structures previously known only in physical contexts. This connection suggests that methods from mathematical physics may yield further insights into financial modeling, particularly for understanding the constraints and symmetries inherent in arbitrage-free pricing.

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